Architectural Style Justification

Architecture Selection: **We chose Microservices Architecture for the Community Lost & Found Matching Network System.**

Architectural Choice

The system is decomposed into six independently deployable services: Listing Service (post intake), Tagging Service (automated visual/text tag extraction), Matching Service (geo-proximity and category matching), Verification Service (proof-of-ownership checks), Notification Service (match alerts), and Admin Service (moderation). Each communicates with the others only through well-defined APIs, as shown in the accompanying component diagram.

Reason 1 — Independent, Uneven Scaling Needs

The services in this system have very different load profiles. Tagging (image/text recognition) and Matching (geo-proximity search across active listings) are computationally heavy and must scale to meet NFR-002's 3-second response target as the number of community listings grows into the thousands. Posting a listing or sending a notification is comparatively lightweight. Microservices let us scale only the Matching and Tagging services horizontally during peak load (e.g., after a large public event generates a surge of lost/found reports), instead of scaling an entire monolith.

Reason 2 — Independent Development and Deployment

Trust-and-safety rules in the Admin Service (what counts as a suspicious listing or match) are expected to change frequently as the community grows, while the core posting and matching flow should remain stable. A microservices approach lets the Admin Service be updated and redeployed on its own release cycle without touching or re-testing the Listing or Matching services, reducing regression risk across unrelated features.

Security Advantage

Proof-of-ownership questions and answers (NFR-001) are isolated inside the Verification Service's own data store, separate from the Listing and Notification services. A vulnerability or bug in a lower-sensitivity service (e.g., Notification Service) cannot directly expose security answers or a finder's private contact details, since those only ever live behind the Verification Service's own API boundary.

Performance Benefit

Because the Matching Service is a separate deployable unit, additional instances can be spun up behind a load balancer specifically for matching queries during high-traffic periods, directly supporting the NFR-002 requirement of returning results in under 3 seconds across 10,000+ active listings — without over-provisioning lighter-weight services like Notification or Admin that do not face the same load.